

COMPASS Detailed Instructions — ASSIGN Grandchild Prototype

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Task: Add and test an experimental Grandchild ASSIGN model while preserving the Parent and Child models unchanged.

1. Mission

We need a minimal, domain-general ASSIGN mechanism for capacity-limited pools (player → team, officer → unit, academic → institution/department, etc.). This is **ASSIGN only**; do not redesign SCORE or SELECT.

Add the Grandchild as a clearly named experimental arm. Do not overwrite, refactor away, or silently alter the Parent or Child.

Scientific question:

Can initially identical teams differentiate endogenously when each arriving individual is attracted by similarity to the team’s current membership centroid, subject to fixed roster capacity?

2. Ingredients and provenance

Configuration/stub logic: $R_j^{(s)}$ is our notation for remaining roster positions / unmatched team-side stubs at assignment step s . Structural lineage: configuration-model stub matching (Fosdick et al., 2018).

Quayle-style homophily: use the scalar exponential distance kernel:

$$e^{-\rho|A_i-\mu_j|}.$$

Here ρ is homophilic preference strength. Conceptual lineage: Quayle, Siddiqui & Jones (2006).

The complete weighted-stub probability is **our synthesis**; do not attribute the combined equation wholesale to either source.

3. The three models

Parent — dynamic constrained homophilic bipartite formation

Individuals have $A_{i,t}$; teams have lagged $\mu_{j,t-1}$. During period t , the centroid is frozen:

$$P(i \rightarrow j \mid t, s) = \frac{R_{j,t}^{(s)} e^{-\rho|A_{i,t}^*-\mu_{j,t-1}|}}{\sum_l R_{l,t}^{(s)} e^{-\rho|A_{i,t}^*-\mu_{l,t-1}|}}.$$

After assignment to j :

$$R_{j,t}^{(s+1)} = R_{j,t}^{(s)} - 1.$$

After the period, a new $\mu_{j,t}$ can govern the next period:

$$\mu_{j,t-1} \rightarrow \text{ASSIGN at } t \rightarrow \mu_{j,t} \rightarrow \text{ASSIGN at } t + 1.$$

This is the general future-facing model for turnover, tenure, reassignment, attrition, and steady-state work. Do not implement those extensions now.

Child — fixed-centroid one-shot generator

Inputs are fixed empirical μ_j , individual A_i , uniform capacity C , and ρ .

Initialize:

$$R_j^{(0)} = C \quad \forall j.$$

Randomly permute players and assign:

$$P(i \rightarrow j \mid s) = \frac{R_j^{(s)} e^{-\rho|A_i-\mu_j|}}{\sum_l R_l^{(s)} e^{-\rho|A_i-\mu_l|}}.$$

The empirical μ_j values remain frozen for the complete formation run.

one formation run = one stochastic realization.

Repeated runs are independent realizations, not longitudinal periods.

Grandchild — endogenous-centroid one-shot generator

Code this model now.

Retain bipartite assignment, hard capacity, R_j , random player order, exponential homophily, and one formation pass.

Change the centroid assumption. All teams begin identical:

$$\mu_j^{(0)} = \mu_0 = \bar{A} \quad \forall j$$

where:

$$\bar{A} = \frac{1}{N} \sum_i A_i.$$

All teams begin empty:

$$R_j^{(0)} = C \quad \forall j.$$

At step s :

$$P(i \rightarrow j \mid s) = \frac{R_j^{(s)} e^{-\rho |A_i - \mu_j^{(s)}|}}{\sum_l R_l^{(s)} e^{-\rho |A_i - \mu_l^{(s)}|}}.$$

After i joins j , decrement R_j , then immediately update the receiving team's centroid from actual members. If it previously contained $n_j^{(s)} > 0$ members:

$$\mu_j^{(s+1)} = \frac{n_j^{(s)} \mu_j^{(s)} + A_i}{n_j^{(s)} + 1}.$$

First-member rule: μ_0 is a starting signal, not a pseudo-player. When the first player attaches to an empty team:

$$\mu_j = A_i.$$

Only the receiving team's state changes at that event.

4. Why test the Grandchild?

The Child's fixed empirical centroids may not generate enough endogenous sorting. The Grandchild begins with **zero between-team heterogeneity**. Any final differentiation must emerge from:

1. stochastic early assignments;
2. immediate centroid updating;
3. similarity-biased later assignments;
4. finite capacity.

Using $\mu_0 = \bar{A}$ creates no initial team advantage or differentiation.

5. Prototype algorithm

For one realization at chosen ρ :

1. Load/construct the existing empirical player ability vector $\{A_i\}$ where practical.
 2. Use a full-capacity test with $N = \sum_j C_j$.
 3. Initial MBB capacity: $C_j = C = 15$.
 4. Compute $\mu_0 = \bar{A}$.
 5. Initialize every team with $R_j = C$, $n_j = 0$, $\mu_j = \mu_0$.
 6. Uniformly randomize player order; never sort by A_i .
 7. For player i , consider all teams with $R_j > 0$.
 8. Compute $w_{ij} = R_j e^{-\rho |A_i - \mu_j|}$.
 9. Normalize $p_{ij} = w_{ij} / \sum_l w_{il}$.
 10. Sample team j stochastically.
 11. Attach i to j .
 12. Set $R_j \leftarrow R_j - 1$.
 13. Immediately update that team's centroid from actual members; first member implies $\mu_j = A_i$.
 14. Continue until all players are assigned and intended slots filled.
 15. Save assignments, final centroids, seed, and diagnostics.
- Each independent realization resets $\{A_i\}$, C , ρ , μ_0 , team states, and player queue. No final state carries forward.

6. Rho sweep

Characterize:

$$\rho \longrightarrow \text{realized compositional sorting.}$$

Begin at $\rho = 0$ and sweep upward over a range broad enough to reveal shape/saturation. Existing project sweep conventions may be reused, but do not assume the old range is optimal.
Run multiple independent realizations per ρ with reproducible seeds and distinct random streams.

Do not assume linearity. The exponential kernel, path dependence, and hard capacity make nonlinear/saturating behavior plausible.

7. Outcome diagnostics

Within-team MSE

For final team mapping $g(i)$:

$$D = \frac{1}{N} \sum_i (A_i - \mu_{g(i)}^{\text{final}})^2.$$

Call this **within-team MSE/dispersion**, not assortativity.
Hypothesized direction:

$$\rho \uparrow \Rightarrow D \downarrow.$$

Test it; do not impose it.

Normalized sorting index

Also calculate:

$$H = 1 - \frac{\sum_i (A_i - \mu_{g(i)}^{\text{final}})^2}{\sum_i (A_i - \bar{A})^2}.$$

This is an explained-variance-style partition statistic. Hypothesized direction:

$$\rho \uparrow \Rightarrow H \uparrow.$$

Also retain final μ_j distribution, SD/variance of final team centroids, within-team distributions, roster sizes, assignment order, and seed.

8. Required plots

Produce at minimum:

1. D vs. ρ : ensemble mean plus variation.
2. H vs. ρ : ensemble mean plus variation.
3. Final team-centroid dispersion vs. ρ .
4. Representative final centroid/team-composition distributions at low, intermediate, and high ρ .

Do not begin by fitting a linear relationship. Visualize the empirical curve first.

9. Validation locks

Assert/check:

- every player has exactly one team;
- no team exceeds C ;
- full-capacity runs finish with every team at exactly C ;
- $\sum_j R_j = 0$ at completion;
- probabilities sum to 1 at every event;
- full teams have zero assignment probability;
- at $\rho = 0$:

$$P(i \rightarrow j \mid s) = \frac{R_j^{(s)}}{\sum_l R_l^{(s)}};$$

- initially all $\mu_j = \bar{A}$;
- first player on an empty team makes $\mu_j = A_i$ exactly;

- μ_0 is never counted as an observation;
 - repeated realizations reset all states;
 - Parent and Child code paths/results remain unchanged.
- Add lightweight assertions/tests.

10. Terminology locks

- A_i : individual ability/performance signal.
- μ_j : team/pool centroid.
- C : team capacity.
- R_j : remaining capacity/unmatched team stubs.
- ρ : homophilic preference strength.
- s : sequential assignment event within one realization.
- t : longitudinal period in the Parent only.
- Homophily $\neq$ realized assortativity/sorting.
- One formation run = one stochastic realization.
- Repeated realizations $\neq$ longitudinal iterations.

11. Do not do these things

Do not modify SCORE or SELECT; add prestige/popularity/preferential attachment; add tenure/turnover; add heterogeneous capacities; carry state between realizations; use empirical team μ_j in the Grandchild; sort players by A_i ; use Quayle's analytic homophily-to-assortativity mapping; call D assortativity; delete/overwrite Parent or Child; or turn this prototype into a large refactor.

12. Deliverables from COMPASS

- Return:
1. Grandchild implementation;
 2. short README/method note stating what was added and what remained untouched;
 3. validation/test output;
 4. ρ -sweep plots;
 5. compact results table by ρ with realization count, mean/SD of D , mean/SD of H , and mean/SD final-centroid dispersion;
 6. short interpretation of observed curve shapes without overclaiming;
 7. any numerical/pathological issues.

13. Recent VECTOR/User Documents to Attach

These are the recent task-specific documents COMPASS may not have in the repo.

VECTOR_ASSIGN_Dynamic_to_OneShot_Model.pptx - this can be found in the 3-Master_Plan/VECTOR_work/VECTOR_ASSIGN_Dynamic_to_OneShot_Model folder of slide images

Two-slide briefing of Parent and Child; configuration/stub component, Quayle component, and our combined synthesis.

VECTOR_One_Page_Homophilic_Initial_League_Model.md - this can be found in the 3-Master_Plan/VECTOR_work folder

Clean Parent-to-Child derivation, notation, fixed empirical-centroid Child, and the corrected identity:

$$\text{one formation run} = \text{one stochastic realization.}$$

VECTOR_Bipartite_Assortative_Formation_Model.md - this can be found in the 3-Master_Plan/VECTOR_work folder

Broader domain-general working specification. It contains the earlier frozen-centroid architecture, capacity, notation, and future longitudinal concepts. Read as background; **the Grandchild intentionally changes the frozen-centroid assumption within a one-shot run.**

Helpful optional Quayle context

If useful, attach **Quayle_One_Page_Formula_Concept_Sheet.md** and/or **Quayle_Companion_Volume.md** for the distinction between similarity, homophily, stochastic attachment, and realized assortativity.

14. Repo / Previously Supplied Context COMPASS Already Has

Because COMPASS has Cursor access to the repo, inspect these first rather than asking for duplicate uploads.

tier1_pool_assignment.py

Start here for current ASSIGN implementation, interfaces, one-shot league generation, randomization, seeds, and code paths the Grandchild should preserve.

`empirical_rho_coverage_overlay.py`

Use for existing ρ -sweep architecture, experiment harness, plotting conventions, and reusable diagnostics.

`09_PD18_Alex_meeting_takeaways.md`

Use for project-level separation of ASSIGN from SCORE/SELECT and the role of ρ as the ASSIGN-side knob.

`20260807_VECTOR_PD18_literature_brief.md`

Use for the PD18 problem statement, empirical motivation, constraints, and context that initiated the ASSIGN modeling search.

Existing Quayle source/material

Use the Quayle paper and any existing project notes for the original homophily mechanism: homophily is a **generative attachment preference**; realized assortativity is an **emergent outcome**.

Do not reopen the broad Saracco/Bomiriya search unless a concrete implementation problem requires it.

15. Success criterion

The prototype succeeds if it reproducibly tells us whether:

initially identical teams + sequential centroid updating + homophilic attachment + hard capacity

produces increasing endogenous compositional differentiation as ρ increases, and what the empirical shape of that relationship is.

We are testing the mechanism, not yet declaring it the final ASSIGN model.